[Oct-25]

GITAM (Deemed to be University)
[CSCI6131]
MCA Degree Examinations

I Semester (Sessional –I)

OPERATING SYSTEMS: CONCEPTS AND DESIGN

(Effective from 2025–26 onwards)

Time: 1 ½ Hour **Max. Marks: 30**

Instruction: All parts of the sections must be answered in one place only

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Section-A

Answer all the following **(2×2 = 4M)**

1. How do sockets facilitate communication between a client and a server?
2. Differentiate between Response Time and Throughput in CPU Scheduling.

Section-B

Answer the following **(2×5 = 10M)**

3. Differentiate between multiprogramming and Distributed operating systems.

OR

4. List and explain system calls that support communication between processes.
5. In a multi-programmed system, demonstrate how long-term scheduling and short-term scheduling work together to improve system performance.

OR

6. Analyze how the benefits of multithreaded programming contribute to improved system performance.

Section-C

Answer the following **(2×8 = 16M)**

7. Elaborate on how an operating system operates in dual modes and illustrate the steps involved in mode switching.

OR

8. Discuss operating system services and explain their significance.

9. Analyze the contents of PCB and explain how they support process execution and scheduling.

OR

10. The following processes are being scheduled using FCFS and Round Robin(RR) scheduling algorithm.

Process	Priority	Burst	Arrival
P1	8	15	0
P2	3	20	0
P3	4	20	20
P4	4	20	25
P5	5	5	45
P6	5	15	55

Given the Time Quanta for RR is 10.

a) Show the scheduling order of the processes using a Gantt chart.
b) What is the turnaround time for each process?
c) What is the waiting time for each process?